Determinant and Diagonalization

Goals

Study about determinant of a matrix and diagonalization a square matrix with application to linear recurrences

1. Compute determinant of a matrix
2. Properties of determinant
3. Use determinant to find inverse of a matrix
4. Use determinant to solve system of linear equations: Cramer's rule
5. Diagonalize a square matrix
6. Application to linear recurrences

Table of contents

1. Determinant
   - Cofactor Expansion
   - Elementary operations
   - Adjugates and Inverse Matrix
   - Cramer's rule
2. Diagonalization
   - Eigenvalues and Eigenvectors
   - Diagonalization
3. An Application to Linear Recurrences
4. Application in Linear Dynamical System

Determinant of a 2×2 matrix

- Determinant of a square matrix is denoted by $\det A$ or $\det(A)$ or $|A|$.
- For a 2×2 matrix

$$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$$

or

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Example

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$$

then $\det(A) = 1 * 5 - 3 * 5 = -10$

In order to define the determinant of 3×3, 4×4 ... matrix with inductive method, we need some terminology

Definition (Co-factor)

Let A be a $n \times n$ matrix and

$(i, j)-$ minor A_{ij} : matrix obtained from A by deleting row i and column j

Then the (i, j) - **cofactor** $c_{ij}(A)$ is the scale defined by

$$c_{ij}(A) = (-1)^{i+j} \det(A_{ij})$$

$(-1)^{i+j}$ is called the **sign** of (i, j) - position

Matrix for sign of position

$$\begin{bmatrix} + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

The signs alternate along each row and column with $+$ in the upper left corner

Example

Find the cofactors of positions (1, 2), (3, 1), and (2, 3) in the following matrix.

$$A = \begin{bmatrix} 3 & -1 & 6 \\ 5 & 2 & 7 \\ 8 & 9 & 4 \end{bmatrix}$$

Example

Find the cofactors of positions (1, 2), (3, 1), and (2, 3) in the following matrix.

$$A = \begin{bmatrix} 3 & -1 & 6 \\ 5 & 2 & 7 \\ 8 & 9 & 4 \end{bmatrix}$$

Solution

In order to compute $c_{12}(A)$, first we delete row 1 and column 2 to get

$$A_{12} = \begin{bmatrix} 5 & 7 \\ 8 & 4 \end{bmatrix}$$

The sign of position $(1, 2)$ is $(-1)^{1+2}$. So the $(1, 2)$ - cofactor

$$c_{12}(A) = (-1)^{1+2} \begin{vmatrix} 5 & 7 \\ 8 & 4 \end{vmatrix} = -(5 * 4 - 7 * 8) = 36$$

Determinant by Cofactor Expansion

Cofactor expansion along row i

$$a_{i1}c_{i1}(A) + a_{i2}c_{i2})(A) + \cdots + a_{in}c_{in}(A)$$

Cofactor expansion along column j

$$a_{1j}c_{1j}(A) + a_{2j}c_{2j})(A) + \cdots + a_{nj}c_{nj}(A)$$

The cofactor expansion along *any row or column* of a matrix A always gives the same result. This result is called the determinant of A and denoted by $\det(A)$ or $\det A$ or $|A|$

Example

Compute the determinant of

$$A = \begin{bmatrix} 3 & 4 & 5 \\ 1 & 7 & 2 \\ 9 & 8 & -6 \end{bmatrix}$$

Solution

The cofactor expansion along the first row

$$\begin{aligned} \det A &= 3c_{11}(A) + 4c_{12}(A) + 5c_{13}(A) \\ &= 3\begin{vmatrix} 7 & 2 \\ 8 & -6 \end{vmatrix} - 4\begin{vmatrix} 1 & 2 \\ 9 & -6 \end{vmatrix} + 3\begin{vmatrix} 1 & 7 \\ 9 & 8 \end{vmatrix} \\ &= 3(-58) - 4(-24) + 5(-55) \\ &= -353 \end{aligned}$$

Compute $\det A$ by expanding along the first column

$$\begin{aligned}\det A &= 3c_{11}(A) + 1c_{21}(A) + 9c_{31}(A)\\ &= 3\begin{vmatrix} 7 & 2 \\ 8 & -6 \end{vmatrix} - \begin{vmatrix} 4 & 5 \\ 8 & -6 \end{vmatrix} + 9\begin{vmatrix} 4 & 5 \\ 7 & 2 \end{vmatrix}\\ &= 3(-58) - (-64) + 9(-27)\\ &= -353\end{aligned}$$

Example

Compute determinant of

$$A = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 5 & 1 & 2 & 0 \\ 2 & 6 & 0 & -1 \\ -6 & 3 & 1 & 0 \end{bmatrix}$$

Solution

The expansion involves multiplying entries by cofactors, so the work is minimized when the row or column contains as many zero entries as possible. Row 1 is a best choice in this matrix. The expansion is

$$\begin{aligned} \det A &= 3c_{11}(A) + 0c_{12}(A) + 0c_{13}(A) + 0c_{14}(A) \\ &= 3\begin{vmatrix} 1 & 2 & 0 \\ 6 & 0 & -1 \\ 3 & 1 & 0 \end{vmatrix} \end{aligned}$$

Solution (cont)

Use expansion along the first row to compute the determinant of 3×3 *matrix*

$$\begin{aligned}\det A &= 3\left(0\begin{vmatrix} 6 & 0 \\ 3 & 1 \end{vmatrix} - (-1)\begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} + 0\begin{vmatrix} 1 & 2 \\ 6 & 0 \end{vmatrix}\right) \\ &= 3[0 + 1(-5) + 0] \\ &= -15\end{aligned}$$

Solution (cont)

Use expansion along the first row to compute the determinant of 3×3 *matrix*

$$\begin{aligned}\det A &= 3\left(0\begin{vmatrix} 6 & 0 \\ 3 & 1 \end{vmatrix} - (-1)\begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} + 0\begin{vmatrix} 1 & 2 \\ 6 & 0 \end{vmatrix}\right) \\ &= 3[0 + 1(-5) + 0] \\ &= -15\end{aligned}$$

It'd better to choose the column 3

Determinant of triangular matrices

Upper Triangular Matrix

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} & a_{23} & \cdots & a_{2n} \\ 0 & 0 & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

Lower Triangular Matrix

$$\begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ a_{21} & a_{22} & 0 & \cdots & 0 \\ a_{31} & a_{32} & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix}$$

Determinant of a triangular matrix is the product of the entries on the main onal.

$$determinant = a_{11}a_{22}\ldots a_{nn}$$

Theorem

Consider matrices $\begin{bmatrix} A & X \\ 0 & B \end{bmatrix}$ *and* $\begin{bmatrix} A & 0 \\ Y & B \end{bmatrix}$ *where* A *and* B *are square matrices. Then*

$$\det \begin{bmatrix} A & X \\ 0 & B \end{bmatrix} = \det(A)\det(B) \quad \textit{and} \quad \det \begin{bmatrix} A & 0 \\ Y & B \end{bmatrix} = \det(A)\det(B)$$

Proof.

Use induction on size k of matrix A and cofactor expansion along the first column for upper triangular case and along last column for lower triangular case. □

Example

$$\det \begin{bmatrix} 2 & 3 & 1 & 3 \\ 1 & -2 & -1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 4 & 1 \end{bmatrix} = \det \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \det \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix} = (1)(-3) = -3$$

Geometric Meaning of Determinant

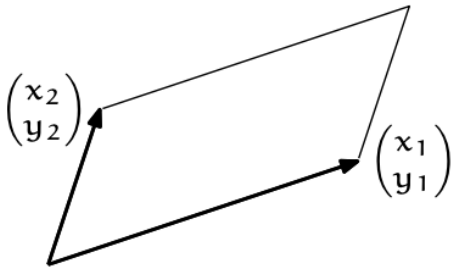
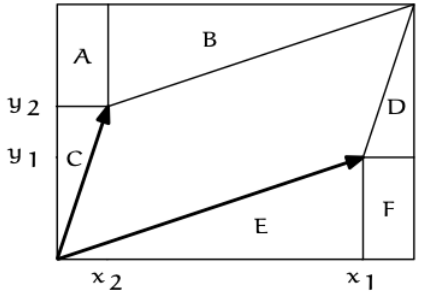


Figure 1: Parallelogram defined by the two vectors



Area of parallelogram defined by two vectors is

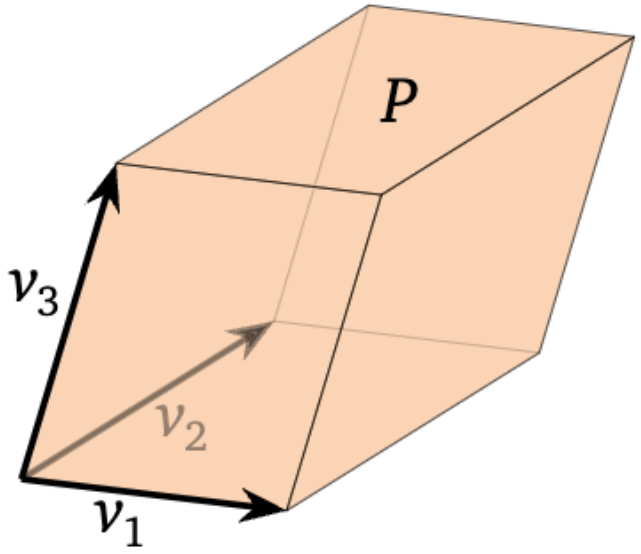
Area of parallelogram =rectangle area − area of A − · · ·

$$=(x_1+x_2)(y_1+y_2)=x_2y_1-x_1y_1/2=x_1y_2-x_2y_1$$

$$=\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix}$$

determinant of the matrix's columns (vectors)

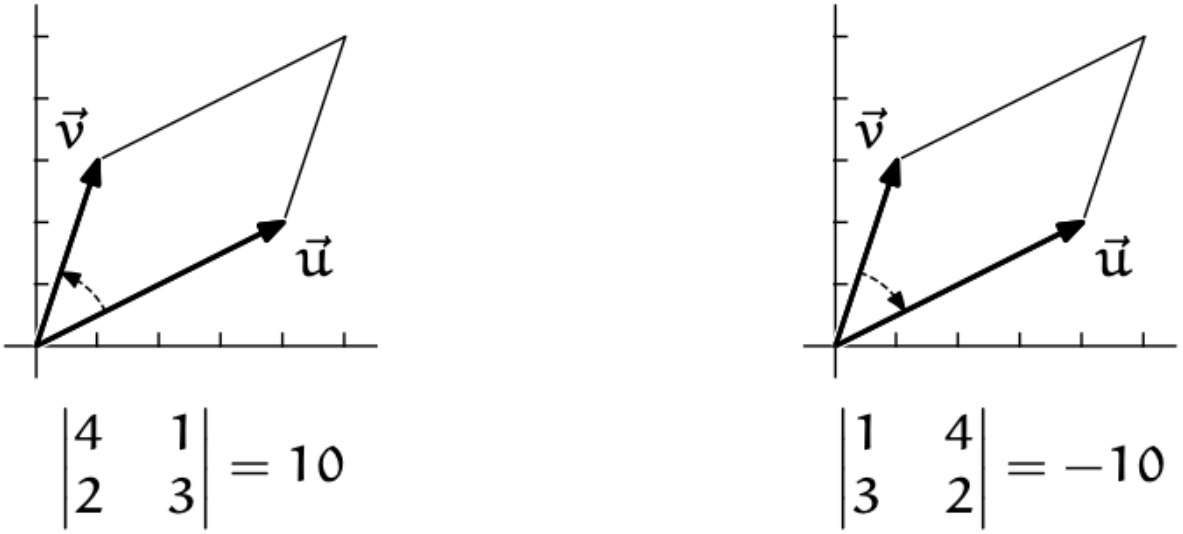
Geometric Meaning of Determinant (cont)



(Oriented) **volume of the parallelepiped** P defined three vector $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$

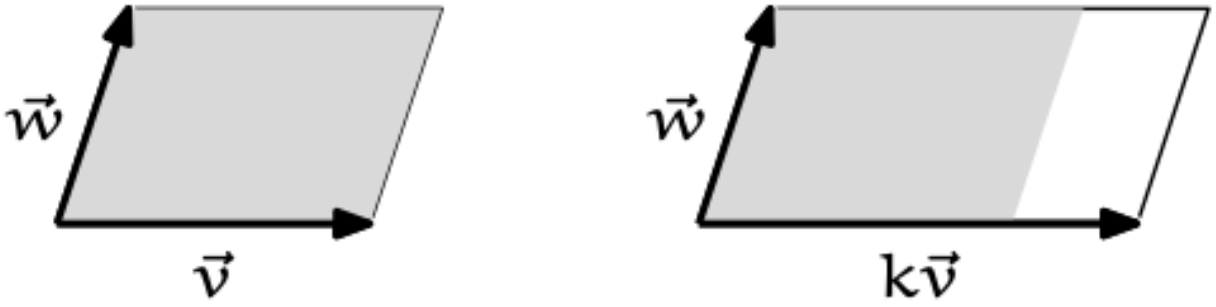
$$vol(P) = \det(\begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix})$$

Orientation - swapping changes the sign of the determinant



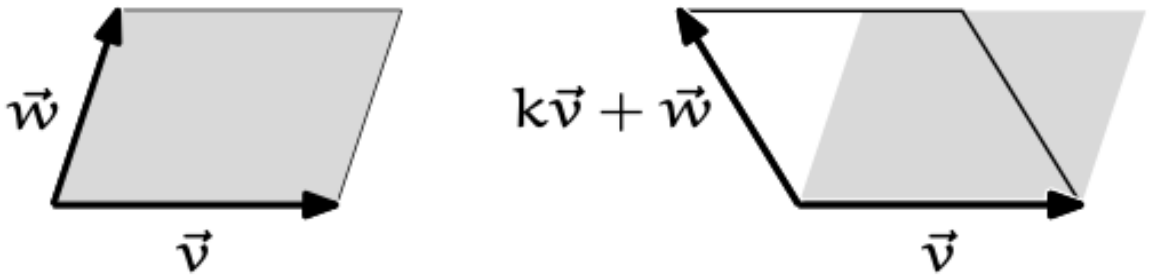
On the left, $\overrightarrow{u}$ is the first column in the matrix and we get a positive size. On the right, $\overrightarrow{v}$ is first and we get a negative size. On the left, the arc from first vector to second is counterclockwise while on the right, the arc from first to second is clockwise. The sign returned by the determinant reflects the orientation or sense of the box.

Scaling effect



If we scale a column by a factor k then the size of the box scales by that factor

Determinant is Unaffected by Combining Columns



The box formed by two vectors $\overrightarrow{v}$ and $k\overrightarrow{v} + \overrightarrow{w}$ is slanted at a different angle than the one form by $\overrightarrow{v}$ and $\overrightarrow{w}$ but two boxes have the same base and height, and hence the same area

Cut down the work to compute determinant

- If A is a 4×4 matrix, the cofactor expansion along any row or column involves calculating four cofactors, each of which involves the determinant of a 3×3 matrix. And if A is 5×5, the expansion involves five determinants of 4×4 matrices

Cut down the work to compute determinant

- If A is a 4×4 matrix, the cofactor expansion along any row or column involves calculating four cofactors, each of which involves the determinant of a 3×3 matrix. And if A is 5×5, the expansion involves five determinants of 4×4 matrices $\Longrightarrow$ *tedious work*

Cut down the work to compute determinant

- If A is a 4×4 matrix, the cofactor expansion along any row or column involves calculating four cofactors, each of which involves the determinant of a 3×3 matrix. And if A is 5×5, the expansion involves five determinants of 4×4 matrices $\Longrightarrow$ *tedious work*
- calculating a determinant is simplified a great deal when a row or column consists mostly of zeros

Cut down the work to compute determinant

- If A is a 4×4 matrix, the cofactor expansion along any row or column involves calculating four cofactors, each of which involves the determinant of a 3×3 matrix. And if A is 5×5, the expansion involves five determinants of 4×4 matrices $\Longrightarrow$ *tedious work*
- calculating a determinant is simplified a great deal when a row or column consists mostly of zeros
- one method of *creating zeros* in a matrix is to apply elementary row operations to it

Properties of Determinant vs Elementary Operations

Let A be a square matrix

1. If A has a row or column of zeros, $\det A = 0$.
   expand along the this row/column
2. If two distinct rows (or columns) of A are interchanged, the determinant of the resulting matrix is $-\det A$.
   Induction
3. If a row (or column) of A is multiplied by a constant u, the determinant of the resulting matrix is $u \det A$.
   expand along this row
4. If two distinct rows (or columns) of A are identical, $\det A = 0$.
   use result in part 2
5. If a multiple of one row of A is added to a different row (or if a multiple of a column is added to a different column), the determinant of the resulting matrix is $\det A$.
   expand along this row (or column) and use the result in part 2

Example

$$\begin{vmatrix} 3 & -1 & 2 \\ 2 & 5 & 1 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

(because the last row consists of zeros)

$$\begin{vmatrix} 3 & -1 & 5 \\ 2 & 8 & 7 \\ 1 & 2 & -1 \end{vmatrix} = -\begin{vmatrix} 5 & -1 & 3 \\ 7 & 8 & 2 \\ -1 & 2 & 1 \end{vmatrix}$$

(because two columns are interchanged)

$$\begin{vmatrix} 8 & 1 & 2 \\ 3 & 0 & 9 \\ 1 & 2 & -1 \end{vmatrix} = 3\begin{vmatrix} 8 & 1 & 2 \\ 1 & 0 & 3 \\ 1 & 2 & -1 \end{vmatrix}$$

(because the second row of the matrix on the left is 3 times the second row of the matrix on the right)

$$\begin{vmatrix} 2 & 1 & 2 \\ 4 & 0 & 4 \\ 1 & 3 & 1 \end{vmatrix} = 0$$

(because two columns are identical)

$$\begin{vmatrix} 2 & 5 & 2 \\ -1 & 2 & 9 \\ 3 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 9 & 20 \\ -1 & 2 & 9 \\ 3 & 1 & 1 \end{vmatrix}$$

(because twice the second row of the matrix on the left was added to the first row)

Example

Evaluate the determinant of

$$A = \begin{bmatrix} 1 & -1 & 3 \\ 1 & 0 & -1 \\ 2 & 1 & 6 \end{bmatrix}$$

Solution

The matrix does have zero entries, so expansion along (say) the second row would involve somewhat less work. However, a column operation can be used to get a zero in position (2, 3) - namely, add column 1 to column 3. Because this does not change the value of the determinant, we obtain

$$\det A = \begin{vmatrix} 1 & -1 & 4 \\ 1 & 0 & 0 \\ 2 & 1 & 8 \end{vmatrix} = -\begin{vmatrix} -1 & 4 \\ 1 & 8 \end{vmatrix} = 12$$

Example

If $\begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix} = 6$, evaluate $\det A$ where

$$A = \begin{bmatrix} a+x & b+y & c+z \\ 3x & 3y & 3z \\ -p & -q & -r \end{bmatrix}$$

Solution

Take common factors out of rows 2 and 3

$$\det A = 3(-1)\begin{vmatrix} a+x & b+y & c+z \\ x & y & z \\ p & q & r \end{vmatrix}$$

Solution (cont)

Subtract the second row from the first

$$\det A = -3 \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}$$

Interchange the last two rows.

$$\det A = 3 \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix} = 3 * 6 = 18$$

Example

Let

$$A = \begin{bmatrix} 1 & x & x \\ x & 1 & x \\ x & x & 1 \end{bmatrix}$$

Find the values of x for which $\det A = 0$

Solution

$$\det A \overset{r_2 - xr_1}{=} \begin{vmatrix} 1 & x & x \\ 0 & 1-x^2 & x-x^2 \\ x & x & 1 \end{vmatrix} \overset{r_3 - xr_1}{=} \begin{vmatrix} 1 & x & x \\ 0 & 1-x^2 & x-x^2 \\ 0 & x-x^2 & 1-x^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1-x^2 & x-x^2 \\ x-x^2 & 1-x^2 \end{vmatrix}$$

Solution (cont)

Taking a common factor of $(1-x)$ *from each row*

$$\det A = \begin{vmatrix} (1-x)(1+x) & x(1-x) \\ x(1-x) & (1-x)(1+x) \end{vmatrix} = (1-x)^2 \begin{vmatrix} 1+x & x \\ x & 1+x \end{vmatrix}$$
$$= (1-x)^2(2x+1)$$

Hence $\det A = 0$ *means* $(1-x)^2(2x+1) = 0$,*that is* $x = 1$ *or* $x = -\frac{1}{2}$.

Exercise

Show that

$$\det \begin{bmatrix} 1 & a_1 & a_1^2 \\ 1 & a_2 & a_2^2 \\ 1 & a_3 & a_3^2 \end{bmatrix} = (a_1 - a_2)(a_2 - a_3)(a_3 - a_1)$$

Properties of Determinant

1. If A is a $n \times n$ matrix then $\det(uA) = u^n A$
   Take the common factor u out of each row and use property 3 in element row operations
2. $\det(AB) = \det(A)\det(B)$
3. $\det(A_1 \dots A_k) = \det(A_1) \dots \det(A_k)$
4. $\det(A^k) = (\det(A))^k$
5. If A is invertible then $\det(A^{-1}) = \frac{1}{\det(A)}$

Inversely if $\det(A) \neq 0$ then A is invertible
In fact, A can be carried to reduced row-echelon matrix R by sequence of elementary matrices $E_1, \ldots, E_k$. That is

$$R = E_k \ldots E_1 A.$$

Since $\det(E) \neq 0$ for all elementary matrix E and $\det(A) \neq 0$, we have

$$\det(R) = \det(E_1) \ldots \det(E_k) \det(A) \neq 0$$

It implies that the square matrix R of reduced row-echelon form has no zero row and then R must be identity matrix. Hence A is invertible

Condition for the existence of the inverse of a matrix

Matrix A is invertible if and only if $\det A \neq 0$

Example

For which values of c does $A = \begin{bmatrix} -1 & 0 & -c \\ -1 & 3 & 1 \\ 0 & 2c & -4 \end{bmatrix}$ have an inverse?

Example

For which values of c does $A = \begin{bmatrix} -1 & 0 & -c \\ -1 & 3 & 1 \\ 0 & 2c & -4 \end{bmatrix}$ have an inverse?

Solution

$$\det(A) \overset{c_3+3\times c_1}{=} \det \begin{bmatrix} -1 & 0 & c \\ -1 & 3 & 1 \\ 0 & 2c & -4 \end{bmatrix} = \det \begin{bmatrix} -1 & 0 & 0 \\ -1 & 3 & 1-c \\ 0 & 2c & -4 \end{bmatrix}$$

$$\overset{\textit{expanding along row 1}}{=} -\det \begin{bmatrix} 3 & 1-c \\ 2c & -4 \end{bmatrix} = -2(c+2)(c-3)$$

A is invertible if and only if $\det(A) \neq 0$. Hence $c \neq -2$ and $c \neq 3$

Exercise

Show that if a product $A_1 \dots A_k$ is invertible then each matrix A_i is invertible.

Determinant of transpose

$$\det(A^T) = \det(A)$$

We can prove by induction with respect to the size of A

- If $n = 2$ then

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \text{ and } A^T = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix}$$

  thus $\det(A) = \det(A^T) = a_{11}a_{22} - a_{12}a_{21}$
- Suppose tht it holds for any $n \times n$ matrix A
- We prove that it holds for any $n+1 \times n+1$ matrix A. In fact, expand along the first row of A and first column of A^T with remark that $A_{ij} = A^T_{ji}$ are $n \times n$ matrix thus $c_{ij}(A) = c_{ji}(A^T)$, we obtain $\det(A) = \det(A^T)$

Example

If $\det(A) = 2$ and $\det(B) = 5$, calculate $\det(A^3 B^{-1} A^T B^2)$.

Solution

$$\det(A^3 B^{-1} A^T B^2) \underbrace{=}_{\textit{determinant of product}} det(A^3)\det(B^{-1})\det(A^T)\det(B^2)$$

$$\underbrace{=}_{\textit{determinant of power}} (\det(A))^3 \det(B^{-1})\det(A^T)(\det(B))^2$$

$$\underbrace{=}_{\textit{determinant of transpose}} (\det(A))^3 \det(B^{-1})\det(A)(\det(B))^2$$

$$\underbrace{=}_{\textit{determinant of inverse}} (\det(A))^4 \frac{1}{\det(B)}(\det(B))^2$$

$$= 2^4 * \frac{1}{5} * 5^2 = \ldots$$

Example

A square matrix is called **orthogonal** if $A^{-1} = A^T$. What are the possible values of $\det(A)$ if A is orthogonal?

Example

A square matrix is called **orthogonal** if $A^{-1} = A^T$. What are the possible values of $\det(A)$ if A is orthogonal?

Solution

If A is orthogonal then $AA^T = I$. Take determinants to obtain

$$\det(I) = \det(AA^T)$$

So

$$1 = \det(A)\det(A^T) = \det(A)\det(A) = [\det(A)]^2$$

Hence $\det(A) = 1$ *or* $\det(A) = -1$.

Definition (Adjugate matrix)

The adjugate of A, denoted $adj(A)$, is the transpose of the cofactor matrix

$$adj(A) = [c_{ij}(A)]^T$$

Theorem (Adjugate Formula)

If A is a square matrix then

$$A(adj(A)) = (adj(A))A = \det(A)I$$

In particular if $\det(A) \neq 0$ then the inverse of A is given by

$$A^{-1} = \frac{1}{\det A} adj(A)$$

Example

Compute the adjugate of

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 0 & 1 & 5 \\ -2 & -6 & 7 \end{bmatrix}$$

and A^{-1}

Solution

Compute the cofactor matrix

$$\begin{bmatrix} c_{11}(A) & c_{12}(A) & c_{13}(A) \\ c_{21}(A) & c_{22}(A) & c_{23}(A) \\ c_{31}(A) & c_{32}(A) & c_{33}(A) \end{bmatrix} = \begin{bmatrix} \begin{vmatrix} 1 & 5 \\ -6 & 7 \end{vmatrix} & -\begin{vmatrix} 0 & 5 \\ -2 & 7 \end{vmatrix} & \begin{vmatrix} 0 & 1 \\ -2 & -6 \end{vmatrix} \\ -\begin{vmatrix} 3 & -2 \\ -6 & 7 \end{vmatrix} & \begin{vmatrix} 1 & -2 \\ -2 & 7 \end{vmatrix} & -\begin{vmatrix} 1 & 3 \\ -2 & -6 \end{vmatrix} \\ \begin{vmatrix} 3 & -2 \\ 1 & 5 \end{vmatrix} & -\begin{vmatrix} 1 & -2 \\ 0 & 5 \end{vmatrix} & \begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} 37 & -10 & 2 \\ -9 & 3 & 0 \\ 17 & -5 & 1 \end{bmatrix}$$

The adjugate of A is the transpose of this cofactor matrix

$$\operatorname{adj} A = \begin{bmatrix} 37 & -10 & 2 \\ -9 & 3 & 0 \\ 17 & -5 & 1 \end{bmatrix}^T = \begin{bmatrix} 37 & -9 & 17 \\ -10 & 3 & -5 \\ 2 & 0 & 1 \end{bmatrix}$$

The determinant of A

$$\det(A) \overset{r_3+2r_1}{=} \begin{vmatrix} 1 & 3 & -2 \\ 0 & 1 & 5 \\ 0 & 0 & 3 \end{vmatrix} = 3$$

The inverse of A

$$A^{-1} = \frac{1}{\det(A)} adj(A) = 3 \begin{bmatrix} 37 & -9 & 17 \\ -10 & 3 & -5 \\ 2 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 111 & -27 & 51 \\ -30 & 9 & -15 \\ 6 & 0 & 3 \end{bmatrix}$$

Example

Find the $(2,3)$ - entry of A^{-1} if $A = \begin{bmatrix} 2 & 1 & 3 \\ 5 & -7 & 1 \\ 3 & 0 & -6 \end{bmatrix}$

Solution

$$\det(A) = \begin{vmatrix} 2 & 1 & 3 \\ 5 & -7 & 1 \\ 3 & 0 & -6 \end{vmatrix} \overset{c_3+2c_1}{=} \begin{vmatrix} 2 & 1 & 7 \\ 5 & -7 & 11 \\ 3 & 0 & 0 \end{vmatrix} = 3 \begin{vmatrix} 1 & 7 \\ -7 & 11 \end{vmatrix} = 180$$

Since $A^{-1} = \frac{1}{\det(A)} adj(A) = \frac{1}{180}[c_{ij}(A)]^T$, *the* $(2,3)$ *- entry of* A^{-1} *is the* $(3,2)$*- entry of* $\frac{1}{180}[c_{ij}(A)]$ *and equals to*

$$\frac{1}{180} c_{32}(A) = \frac{1}{180}\left(-\begin{vmatrix} 2 & 3 \\ 5 & 1 \end{vmatrix}\right) = \frac{13}{180}$$

Example

Let A be an $n \times n$ matrix, $n \geq 2$. Show that $\det(adj(A)) = (\det(A))^{n-1}$

Example

Let A be an $n \times n$ matrix, $n \geq 2$. Show that $\det(adj(A)) = (\det(A))^{n-1}$

Solution

Denote $d = \det(A)$. *From adjugate formula, we have*

$$\det(Aadj(A)) = \det(dI) = d^n \det(I) = d^n$$

Moreover

$$\det(Aadj(A)) = \det(A)\det(adj(A)) = d \times \det(adj(A))$$

So

$$d \times \det(adj(A)) = d^n$$

If $d \neq 0$ *then we have* $\det(adj(A)) = (\det(A))^{n-1}$.
Inversely if $d = 0$ *then from adjugat formula,* $Aadj(A) = 0$. *If* $\det(adj(A)) \neq 0$ *then* $adj(A)$ *is invertible and* $A = 0(adj(A)^{-1}) = 0$. *It implies that* $adj(A) = 0$ *which is impossible. Hence* $\det(adj(A)) = 0 = d$

Come back to linear system with unique solution

Suppose that

$$A\mathbf{x} = \mathbf{b}$$

is a system of linear equations of n variables $x_1, \ldots, x_n$ with

- A is a $n \times n$ matrix
- $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$

If $\det(A) \neq 0$ then the system solution has unique solution

$$\mathbf{x} = A^{-1}\mathbf{b} \ = \frac{1}{\det(A)}(adj(A))\mathbf{b}$$

that is $\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \frac{1}{\det(A)} \begin{bmatrix} c_{11}(A) & c_{21}(A) & \ldots & c_{nn}(A) \\ c_{12}(A) & c_{22}(A) & \ldots & c_{n2}(A) \\ \vdots & \vdots & \ddots & \vdots \\ c_{1n} & c_{2n} & \ldots & c_{nn} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$

So we have

$$x_1 = \frac{1}{\det(A)}[b_1 c_{11}(A) + b_2 c_{22}(A) + \cdots + b_n c_{n1}(A)]$$

The quantity $b_1 c_{11}(A) + b_2 c_{21}(A) + \cdots + b_n c_{n1}(A)$ occurring in the formula looks like the cofactor expansion of the determinant of a matrix. The cofactors involved are $c_{11}(A), c_{21}(A), ..., c_{n1}(A)$, corresponding to the first column of A. If A_1 is obtained from A by replacing the first column of A by **b** then expanding $\det(A_1)$ by the first column gives

$$\det(A_1) = b_1 c_{11}(A) + b_2 c_{22}(A) + \cdots + b_n c_{n1}(A) = \det(A) x_1$$

Simular result for $x_2, \ldots$

Theorem (Cramer's rule)

If A is invertible $n \times n$ matrix then the solution of the system

$$A\boldsymbol{x} = \boldsymbol{b}$$

of n equations in n variables $x_1, \ldots, x_n$ is given by

$$x_1 = \frac{\det(A_1)}{\det(A)}, \; x_2 = \frac{\det(A_2)}{\det(A)}, \ldots, x_n = \frac{\det(A_n)}{\det(A)}$$

where, for each k, A_k is the matrix obtained from A by replacing column k by $\boldsymbol{b}$.

Example

Find x_1 given the following system of equations

$$5x_1 + x_2 - x_3 = 4$$
$$9x_1 + x_2 - x_3 = 1$$
$$x_1 - x_2 + 5x_3 = 2$$

Solution

$$\det(A) = \begin{vmatrix} 5 & 1 & -1 \\ 9 & 1 & -1 \\ 1 & -1 & 5 \end{vmatrix} = -16$$

$$\det(A_1) = \begin{vmatrix} 4 & 1 & -1 \\ 1 & 1 & -1 \\ 2 & -1 & 5 \end{vmatrix} = 12$$

Hence $x_1 = \frac{\det(A_1)}{\det(A)} = \frac{12}{-16} = -\frac{3}{4}$ *by Cramer's rule.*

Table of contents

1. Determinant
   - Cofactor Expansion
   - Elementary operations
   - Adjugates and Inverse Matrix
   - Cramer's rule
2. Diagonalization
   - Eigenvalues and Eigenvectors
   - Diagonalization
3. An Application to Linear Recurrences
4. Application in Linear Dynamical System

Motivation example

Consider the evolution of the population of a species of birds. Because the number of males and females are nearly equal, we count only females. We assume that each female remains a juvenile for one year and then becomes an adult, and that only adults have offspring. We make three assumptions about reproduction and survival rates:

1. The number of juvenile females hatched in any year is twice the number of adult females alive the year before (we say the reproduction rate is 2).
2. Half of the adult females in any year survive to the next year (the adult survival rate is $\frac{1}{2}$).
3. One quarter of the juvenile females in any year survive into adulthood (the juvenile survival rate is $\frac{1}{4}$) .

If there were 100 adult females and 40 juvenile females alive initially, compute the population of females k years late

- a_k and j_k : the number of adult and juvenile females after k years
- total female population is $a_k + j_k$.
- Assumption 1 shows that $j_{k+1} = 2a_k$
- assumptions 2 and 3 show that $a_{k+1} = \frac{1}{2}a_k + \frac{1}{4}j_k$.

Hence

$$a_{k+1} = \frac{1}{2}a_k + \frac{1}{4}j_k$$

$$j_{k+1} = 2a_k$$

Write $v_k = \begin{bmatrix} a_k \\ j_k \end{bmatrix}$ and $A = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ 2 & 0 \end{bmatrix}$ then

$$v_{k+1} = Av_k \textbf{ matrix recurrence}$$

So $v_1 = Av_0$, $v_2 = Av_1 = A^2v_0$, ... In general

$$v_k = A^k v_0$$

with $v_0 = \begin{bmatrix} 100 \\ 40 \end{bmatrix}$

- Direct computation of the powers A^k can be time-consuming $\rightarrow$ need an indirect method
- the power D^k of a diagonal matrix D are easy to compute

$$D = \begin{bmatrix} d_{11} & 0 & \dots & 0 \\ 0 & d_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_{nn} \end{bmatrix} \Longrightarrow D^k = \begin{bmatrix} d_{11}^k & 0 & \dots & 0 \\ 0 & d_{22}^k & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_{nn}^k \end{bmatrix}$$

Idea: first **diagonalize** the matrix A, that is find an invertible matrix P such that

$$P^{-1}AP = D \text{ is a diagonal matrix}$$

or

$$A = PDP^{-1}$$

then compute A^k

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD^2P^{-1},$$
$$A^3 = AA^2 = (PDP^{-1})(PD^2P^{-1}) = PD^3P^{-1}$$

Theorem

If $A = PDP^{-1}$ *then* $A^k = PD^kP^{-1}$ *for* $k = 1, 2, \ldots$

Theorem

If $A = PDP^{-1}$ *then* $A^k = PD^kP^{-1}$ *for* $k = 1, 2, \ldots$

The process of finding an invertible matrix P and a diagonal matrix D so that $A = PDP^{-1}$ is referred to as diagonalizing the matrix A, and P is called the diagonalizing matrix for A.

Theorem

If $A = PDP^{-1}$ *then* $A^k = PD^kP^{-1}$ *for* $k = 1, 2, \ldots$

The process of finding an invertible matrix P and a diagonal matrix D so that $A = PDP^{-1}$ is referred to as diagonalizing the matrix A, and P is called the diagonalizing matrix for A.

Questions

- When is it possible to diagonalize a matrix?

Theorem

If $A = PDP^{-1}$ *then* $A^k = PD^kP^{-1}$ *for* $k = 1, 2, \ldots$

The process of finding an invertible matrix P and a diagonal matrix D so that $A = PDP^{-1}$ is referred to as diagonalizing the matrix A, and P is called the diagonalizing matrix for A.

Questions

- When is it possible to diagonalize a matrix?
- How do we find a diagonalizing matrix?

Theorem

If $A = PDP^{-1}$ *then* $A^k = PD^kP^{-1}$ *for* $k = 1, 2, \ldots$

The process of finding an invertible matrix P and a diagonal matrix D so that $A = PDP^{-1}$ is referred to as diagonalizing the matrix A, and P is called the diagonalizing matrix for A.

Questions

- When is it possible to diagonalize a matrix?
- How do we find a diagonalizing matrix?

Answer

Eigenvalues and eigenvectors.

Eigenvalues and Eigenvectors

Definition

If A is an $n \times n$ matrix, a number λ is called an **eigenvalue** of A if $A\mathbf{x} = \lambda\mathbf{x}$ for some column $\mathbf{x} \neq 0$ in $\mathbb{R}^n$.
In this case, $\mathbf{x}$ is called an **eigenvector of A corresponding to the eigenvalue** λ , or a λ-eigenvector for short.

Example

If $A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$ then $A\mathbf{x} = 4\mathbf{x}$. So $\lambda = 4$ is an eigenvalue of A corresponding to the eigenvector $\mathbf{x}$.

λ is an eigenvalue of matrix A if and only if $A\mathbf{x} = \lambda\mathbf{x}$ for some $\mathbf{x} \neq 0$. This is equivalent to the system

$$(\lambda I - A)\mathbf{x} \ = 0$$

has non trivial solution. This happens if and only if $\lambda I - A$ is not invertible.

λ is an eigenvalue of matrix A if and only if $A\mathbf{x} = \lambda\mathbf{x}$ for some $\mathbf{x} \neq 0$. This is equivalent to the system

$$(\lambda I - A)\mathbf{x} \ = 0$$

has non trivial solution. This happens if and only if $\lambda I - A$ is not invertible.

Definition (Characteristic Polynomial of a Matrix)

If A is an $n \times n$ matrix then the **characteristic polynomial** of A is

$$c_A(x) = \det(xI - A)$$

Theorem

Let A is an $n \times n$ matrix.

1. *The eigenvalues λ of A are the roots of the characteristic polynomial $c_A(x)$ of A.*
2. *The λ-eigenvectors $\mathbf{x}$ are the nonzero solutions to the homogeneous system*

$$(\lambda I - A)\mathbf{x} = 0$$

Procedure to find eigenvalues and eigenvectors

Let A be an $n \times n$ matrix.

- **Eigenvalues:** Find λ by solving the equation

$$c_A(x) = \det(xI - A) = 0$$

Procedure to find eigenvalues and eigenvectors

Let A be an $n \times n$ matrix.

- **Eigenvalues:** Find λ by solving the equation

$$c_A(x) = \det(xI - A) = 0$$

- **Eigenvectors:** For each λ, find $X \neq 0$ by finding the basic solutions to

$$(A - \lambda I)X = 0$$

Procedure to find eigenvalues and eigenvectors

Let A be an $n \times n$ matrix.

- **Eigenvalues:** Find λ by solving the equation

$$c_A(x) = \det(xI - A) = 0$$

- **Eigenvectors:** For each λ, find $X \neq 0$ by finding the basic solutions to

$$(A - \lambda I)X = 0$$

- **Check:** For each pair of λ, X check that $AX = \lambda X$.

Example

Find the characteristic polynomial of the matrix $A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$ and then find all the eigenvalues and their eigenvectors.

Solution

$$xI - A = \begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix} - \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} x-3 & -5 \\ 1 & x+1 \end{bmatrix}$$

$$c_A(x) = \det \begin{bmatrix} x-3 & 5 \\ 1 & x+1 \end{bmatrix} = x^2 - 2x - 8 = (x-4)(x+2)$$

The roots of $c_A(x)$ are $\lambda_1 = -2$ and $\lambda_2 = 4$ and these are the eigenvalues

Solution (cont)

Find the eigenvectors corresponding to $\lambda_1 = -2$
We have

$$\lambda_1 I - A = \begin{bmatrix} -2-3 & -5 \\ -1 & -2+1 \end{bmatrix} = \begin{bmatrix} -5 & -5 \\ -1 & -1 \end{bmatrix}$$

General solution to $(\lambda_1 - A)\boldsymbol{x} = 0$ *are* $\boldsymbol{x} = \begin{bmatrix} -t \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. *So the eigenvectors corresponding to* $\lambda_1 = -2$ *is* $\boldsymbol{x} = \begin{bmatrix} -t \\ t \end{bmatrix}$ *for* $t \neq 0$.

Similar, the eigenvectors corresponding to $\lambda_2 = 4$ *are* $\boldsymbol{x} = \begin{bmatrix} 5t \\ t \end{bmatrix} = t \begin{bmatrix} 5 \\ 1 \end{bmatrix}$ *for* $t \neq 0$

Basic Eigenvectors

Definition

A basic eigenvector of an $n \times n$ matrix A is any nonzero multiple of a basic solution to $(\lambda \mathbf{I} - A)X = 0$, where λ is an eigenvalue of A.

Basic Eigenvectors

Definition

A basic eigenvector of an $n \times n$ matrix A is any nonzero multiple of a basic solution to $(\lambda \mathbf{I} - A)X = 0$, where λ is an eigenvalue of A.

Basic eigenvectors of $A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$

Basic Eigenvectors

Definition

A basic eigenvector of an $n \times n$ matrix A is any nonzero multiple of a basic solution to $(\lambda \mathbf{I} - A)X = 0$, where λ is an eigenvalue of A.

Basic eigenvectors of $A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$

$X = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ is is a basic eigenvector of A corresponding to the eigenvalue -2.

Basic Eigenvectors

Definition

A basic eigenvector of an $n \times n$ matrix A is any nonzero multiple of a basic solution to $(\lambda \mathbf{I} - A)X = 0$, where λ is an eigenvalue of A.

Basic eigenvectors of $A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$

$X = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ is is a basic eigenvector of A corresponding to the eigenvalue -2. $X = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$ is a basic eigenvector of A corresponding to the eigenvalue 4.

Example

Find the characteristic polynomial, eigenvalues, and basic eigenvectors for

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 3 & -2 \end{bmatrix}$$

Solution

Characteristic polynomial

$$c_A(x) = \det \begin{bmatrix} x-2 & 0 & 0 \\ -1 & x-2 & 1 \\ 11 & -3 & x+2 \end{bmatrix} = (x-2)(x-1)(x+1)$$

so the eigenvalues are $\lambda_1 = 2$, $\lambda_2 = 1$ *and* $\lambda_3 = -1$.

Solution (cont)

To find all the eigenvectors corresponding to $\lambda_1 = 2$, *compute*

$$\lambda_1 \boldsymbol{I} - A = \begin{bmatrix} \lambda_1 - 2 & 0 & 0 \\ -1 & \lambda_1 - 2 & 1 \\ 11 & -3 & \lambda_1 + 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 0 & 1 \\ 11 & -3 & 4 \end{bmatrix}$$

Solve $(\lambda_1 \boldsymbol{I} - A)\boldsymbol{x} = 0$ *by gaussian algorithm to obtain the general solution* $x = t \begin{bmatrix} 1 \\ 1 \\ 11 \end{bmatrix}$. *So we can use* $\boldsymbol{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ *as the basic eigenvector corresponding to* $\lambda_1 = 2$.

Similary $\boldsymbol{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ *and* $\boldsymbol{x}_3 = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$ *are the basic eigenvectors corresponding to* $\lambda_2 = 1$ *and* $\lambda_3 = -1$

Eigenvalues with multiplicity greater than one

Problem

Find the characteristic polynomial and eigenvalues of the matrix

$$A = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{bmatrix}.$$

Eigenvalues with multiplicity greater than one

Problem

Find the characteristic polynomial and eigenvalues of the matrix

$$A = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{bmatrix}.$$

Solution

$$c_A(x) = \det(x\mathbf{I} - A)$$

Eigenvalues with multiplicity greater than one

Problem

Find the characteristic polynomial and eigenvalues of the matrix

$$A = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{bmatrix}.$$

Solution

$$c_A(x) = \det(x\mathbf{I} - A) \quad = \quad \det \begin{bmatrix} x-4 & -1 & -2 \\ 0 & x-3 & 2 \\ 0 & 1 & x-2 \end{bmatrix}$$

Eigenvalues with multiplicity greater than one

Problem

Find the characteristic polynomial and eigenvalues of the matrix

$$A = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{bmatrix}.$$

Solution

$$\begin{aligned} c_A(x) = \det(x\mathbf{I} - A) &= \det \begin{bmatrix} x-4 & -1 & -2 \\ 0 & x-3 & 2 \\ 0 & 1 & x-2 \end{bmatrix} \\ &= (x-4)[(x-3)(x-2)-2] \end{aligned}$$

Eigenvalues with multiplicity greater than one

Problem

Find the characteristic polynomial and eigenvalues of the matrix

$$A = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{bmatrix}.$$

Solution

$$\begin{aligned} c_A(x) = \det(x\mathbf{I} - A) &= \det \begin{bmatrix} x-4 & -1 & -2 \\ 0 & x-3 & 2 \\ 0 & 1 & x-2 \end{bmatrix} \\ &= (x-4)[(x-3)(x-2)-2] \\ &= (x-4)(x^2-5x+4) \end{aligned}$$

Eigenvalues with multiplicity greater than one

Problem

Find the characteristic polynomial and eigenvalues of the matrix

$$A = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{bmatrix}.$$

Solution

$$\begin{aligned} c_A(x) = \det(x\mathbf{I} - A) &= \det \begin{bmatrix} x-4 & -1 & -2 \\ 0 & x-3 & 2 \\ 0 & 1 & x-2 \end{bmatrix} \\ &= (x-4)[(x-3)(x-2)-2] \\ &= (x-4)(x^2-5x+4) \\ &= (x-4)(x-4)(x-1) \end{aligned}$$

Eigenvalues with multiplicity greater than one

Problem

Find the characteristic polynomial and eigenvalues of the matrix

$$A = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{bmatrix}.$$

Solution

$$\begin{aligned} c_A(x) = \det(x\mathbf{I} - A) &= \det \begin{bmatrix} x-4 & -1 & -2 \\ 0 & x-3 & 2 \\ 0 & 1 & x-2 \end{bmatrix} \\ &= (x-4)[(x-3)(x-2)-2] \\ &= (x-4)(x^2-5x+4) \\ &= (x-4)(x-4)(x-1) \\ &= (x-4)^2(x-1). \end{aligned}$$

Definition

The multiplicity of an eigenvalue λ of A is the number of times λ occurs as a root of $c_A(x)$.

Definition

The multiplicity of an eigenvalue λ of A is the number of times λ occurs as a root of $c_A(x)$.

Example

We have seen that $A = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{bmatrix}$ has eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 4$ of multiplicity two.

Example (continued)

To find an eigenvector of A corresponding to $\lambda_1 = 1$, solve the homogeneous system $(\mathbf{I} - A)X = 0$:

Example (continued)

To find an eigenvector of A corresponding to $\lambda_1 = 1$, solve the homogeneous system $(\mathbf{I} - A)X = 0$:

$$\left[\begin{array}{rrr|r} -3 & -1 & -2 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right]$$

Example (continued)

To find an eigenvector of A corresponding to $\lambda_1 = 1$, solve the homogeneous system $(\mathbf{I} - A)X = 0$:

$$\left[\begin{array}{rrr|r} -3 & -1 & -2 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right] \to \left[\begin{array}{rrr|r} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Example (continued)

To find an eigenvector of A corresponding to $\lambda_1 = 1$, solve the homogeneous system $(\mathbf{I} - A)X = 0$:

$$\left[\begin{array}{rrr|r} -3 & -1 & -2 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right] \to \left[\begin{array}{rrr|r} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

The general solution is $X = \begin{bmatrix} -s \\ s \\ s \end{bmatrix}$ where $s \in \mathbb{R}$.

Example (continued)

To find an eigenvector of A corresponding to $\lambda_1 = 1$, solve the homogeneous system $(\mathbf{I} - A)X = 0$:

$$\left[\begin{array}{rrr|r} -3 & -1 & -2 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right] \to \left[\begin{array}{rrr|r} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

The general solution is $X = \begin{bmatrix} -s \\ s \\ s \end{bmatrix}$ where $s \in \mathbb{R}$. We get a basic eigenvector by choosing $s = 1$ (in fact, any nonzero value of s gives us a basic eigenvector).

Exercise

For

$$A = \begin{bmatrix} 3 & -4 & 2 \\ 1 & -2 & 2 \\ 1 & -5 & 5 \end{bmatrix},$$

find $c_A(x)$, the eigenvalues of A, and basic eigenvector(s) for each eigenvalue.

Example (continued)

Therefore, $X = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ is a (basic) eigenvector of A corresponding to $\lambda_1 = 1$.

Example (continued)

Therefore, $X = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ is a (basic) eigenvector of A corresponding to $\lambda_1 = 1$.

To find an eigenvector of $A = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{bmatrix}$ corresponding the $\lambda_2 = 4$, solve the system $(4I - A)X = 0$:

$$\left[\begin{array}{ccc|c} 0 & -1 & -2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 0 \end{array}\right]$$

Example (continued)

Therefore, $X = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ is a (basic) eigenvector of A corresponding to $\lambda_1 = 1$.

To find an eigenvector of $A = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{bmatrix}$ corresponding the $\lambda_2 = 4$, solve the system $(4I - A)X = 0$:

$$\left[\begin{array}{ccc|c} 0 & -1 & -2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Example (continued)

Therefore, $X = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ is a (basic) eigenvector of A corresponding to $\lambda_1 = 1$.

To find an eigenvector of $A = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{bmatrix}$ corresponding the $\lambda_2 = 4$, solve the system $(4I - A)X = 0$:

$$\left[\begin{array}{ccc|c} 0 & -1 & -2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

The general solution is $X = \begin{bmatrix} s \\ -2t \\ t \end{bmatrix}$ where $s, t \in \mathbb{R}$.

Example (continued)

In this case, the general solution has two parameters, which leads to two basic eigenvectors that are not scalar multiples of each other, i.e., since

Example (continued)

In this case, the general solution has two parameters, which leads to two basic eigenvectors that are not scalar multiples of each other, i.e., since

$$X = \begin{bmatrix} s \\ -2t \\ t \end{bmatrix}$$

Example (continued)

In this case, the general solution has two parameters, which leads to two basic eigenvectors that are not scalar multiples of each other, i.e., since

$$X = \begin{bmatrix} s \\ -2t \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} \text{ where } s, t \in \mathbb{R},$$

Example (continued)

In this case, the general solution has two parameters, which leads to two basic eigenvectors that are not scalar multiples of each other, i.e., since

$$X = \begin{bmatrix} s \\ -2t \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} \text{ where } s, t \in \mathbb{R},$$

we obtain basic eigenvectors

$$X_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ and } X_2 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}.$$

Example (continued)

In this case, the general solution has two parameters, which leads to two basic eigenvectors that are not scalar multiples of each other, i.e., since

$$X = \begin{bmatrix} s \\ -2t \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} \text{ where } s, t \in \mathbb{R},$$

we obtain basic eigenvectors

$$X_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ and } X_2 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}.$$

We can obtain other pairs of basic 4-eigenvectors for A by taking any nonzero scalar multiple of X_1, and any nonzero scalar multiple of X_2.

Example (continued)

In this case, the general solution has two parameters, which leads to two basic eigenvectors that are not scalar multiples of each other, i.e., since

$$X = \begin{bmatrix} s \\ -2t \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} \text{ where } s, t \in \mathbb{R},$$

we obtain basic eigenvectors

$$X_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ and } X_2 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}.$$

We can obtain other pairs of basic 4-eigenvectors for A by taking any nonzero scalar multiple of X_1, and any nonzero scalar multiple of X_2.

Notice that every 4-eigenvector of A is a nonzero linear combination of basic 4-eigenvectors.

Diagonal matrix

An $n \times n$ matrix D is called a **diagonal matrix** if all its entries off the main diagonal are zero

$$D = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

Denote $D = diag(\lambda_1, \lambda_2, \dots, \lambda_n)$

Diagonalizable Matrices

An $n \times n$ matrix A is called **diagonalizable** if there exists an invertible matrix P such that

$$P^{-1}AP \text{ is diagonal form}$$

The invertible matrix P is called a **diagonalizing matrix** for A

Denote $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ the columns of P and write $P = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \ldots & \mathbf{x}_n \end{bmatrix}$
Then $P^{-1}AP = D$ for some diagonal matrix D implied that

$$AP = PD$$

which is equivalent to

$$\begin{bmatrix} A\mathbf{x}_1 & A\mathbf{x}_2 & \ldots & A\mathbf{x}_n \end{bmatrix} = \begin{bmatrix} \lambda_1\mathbf{x}_1 & \lambda_2\mathbf{x}_2 & \ldots & \lambda_n\mathbf{x}_n \end{bmatrix}$$

So

$$A\mathbf{x}_k = \lambda_k\mathbf{x}_k$$

for all $k = 1, 2, \ldots, n$.
In other words, $P^{-1}AP = D$ holds if and only if the diagonal entries of diagonal D are eigenvalues of A and the columns of the diagonalizing matrix P are corresponding eigenvectors.

Diagonalization Algorithm

To diagonalize an $n \times n$ matrix A:

Step 1 Find the distinct eigenvalues λ of A.

Step 2 Compute a set of basic eigenvectors corresponding to each of these eigenvalues λ as basic solutions of the homogeneous system $(\lambda I - A)\mathbf{x} = 0$.

Step 3 The matrix A is diagonalizable if and only if there are n basic eigenvectors in all.

Step 4 If A is diagonalizable, the $n \times n$ matrix P with these basic eigenvectors as its columns is a diagonalizing matrix for A, that is, P is invertible and $P^{-1}AP$ is diagonal.

Example

Diagonalize the matrix

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 3 & -2 \end{bmatrix}$$

Solution

The matrix A *has three eigenvalues* $\lambda_1 = 2$, $\lambda_2 = 1$ *and* $\lambda_3 = -1$ *with corresponding basis eigenvectors* $\boldsymbol{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $\boldsymbol{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ *and* $\boldsymbol{x}_3 = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$

The diagonalizing matrix

$$P = \begin{bmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_2 & \boldsymbol{x}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

and

$$P^{-1}AP = D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Table of contents

1. Determinant
   - Cofactor Expansion
   - Elementary operations
   - Adjugates and Inverse Matrix
   - Cramer's rule
2. Diagonalization
   - Eigenvalues and Eigenvectors
   - Diagonalization
3. An Application to Linear Recurrences
4. Application in Linear Dynamical System

Recurrence

It often happens that a problem can be solved by finding a sequence of numbers $x_0, x_1, x_2, \ldots$ where the first few are known, and subsequent numbers are given in terms of earlier ones

Example

Fibonacci sequence

$$f_0 = 1, \quad f_1 = 1, \quad f_n = f_{n-1} + f_{n-2} \text{ for } n \geq 2$$

Example

An urban planner wants to determine the number x_k of ways that a row of k parking spaces can be filled with cars and trucks if trucks take up two spaces each. Find the first few values of x_k.

Recurrence

It often happens that a problem can be solved by finding a sequence of numbers $x_0, x_1, x_2, \ldots$ where the first few are known, and subsequent numbers are given in terms of earlier ones

Example

Fibonacci sequence

$$f_0 = 1, \quad f_1 = 1, \quad f_n = f_{n-1} + f_{n-2} \text{ for } n \geq 2$$

Example

An urban planner wants to determine the number x_k of ways that a row of k parking spaces can be filled with cars and trucks if trucks take up two spaces each. Find the first few values of x_k.

Solution

Some first few terms $x_0 = 1$, $x_1 = 1$, $x_2 = 2$ *and* $x_{k+1} = x_k + x_{k+1} \forall k \geq 0$

Question

How to find explicit formula for terms in a sequence given by a linear recursive relation?

Example

Suppose the number $x_0, x_1 \ldots$ are given by the linear recursive relation

$$x_{k+2} = x_{k+1} + 6x_k$$

with $x_0 = 1, x_1 = 1$.

Solution

The idea is to compute the sequence $\boldsymbol{v}_0, \boldsymbol{v}_1 \ldots$ *of columns instead of the numbers* $x_0, x_1, \ldots$ *where*

$$\boldsymbol{v}_k = \begin{bmatrix} x_k \\ x_{k+1} \end{bmatrix} \text{ for } k \geq 0.$$

We have $\boldsymbol{v}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ *and*

$$\boldsymbol{v}_{k+1} = \begin{bmatrix} x_{k+1} \\ x_{k+2} \end{bmatrix} = \begin{bmatrix} x_{k+1} \\ x_k + x_{k+1} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_k \\ x_{k+1} \end{bmatrix} = A\boldsymbol{v}_k \text{ with } A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

Solution (cont)

It is easy to see that $\mathbf{v}_k = A^k \mathbf{v}_0$. So we first diagonalize A and then compute A^k to obtain the general formula for x_k.

Solution (cont)

It is easy to see that $\mathbf{v}_k = A^k \mathbf{v}_0$. So we first diagonalize A and then compute A^k to obtain the general formula for x_k.

Diagonalize A

The characteric polynomial is

$c_A(x) \equiv det(A - x\mathbf{I}) = \det \begin{bmatrix} -x & 1 \\ 1 & 1-x \end{bmatrix} = x^2 - x - 1$ with roots $\frac{1 \pm \sqrt{5}}{2}$ so A has two eigenvalues

$$\lambda_1 = \frac{1+\sqrt{5}}{2}, \quad \lambda_2 = \frac{1-\sqrt{5}}{2}$$

The corresponding eigenvectors are $\mathbf{x}_1 = \begin{bmatrix} 1 \\ \lambda_1 \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} 1 \\ \lambda_2 \end{bmatrix}$.

The diagonalizing matrix for A is $P = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ \lambda_1 & \lambda_2 \end{bmatrix}$ which is invertible

Solution (cont)

Find general formula of x_k

We have $P^{-1}AP = D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$ So

$$D^k = (P^{-1}AP)(P^{-1}AP)\dots P^{-1}AP = P^{-1}A^kP$$

It implies that

$$A^k = PD^kP^{-1}$$

and then

$$\begin{bmatrix} v_k \\ v_{k+1} \end{bmatrix} = \mathbf{v}_k = A^k\mathbf{v}_0 = PD^kP^-1\mathbf{v}_0 = \begin{bmatrix} \frac{\lambda_1^{k+1}-\lambda_2^{k+1}}{\sqrt{5}} \\ \frac{\lambda_1^{k+2}-\lambda_2^{k+2}}{\sqrt{5}} \end{bmatrix}$$

Comparing top entries give an exact formula for x_k

$$x_k = \frac{\lambda_1^{k+1} - \lambda_2^{k+1}}{\sqrt{5}}$$

Table of contents

1. Determinant
   - Cofactor Expansion
   - Elementary operations
   - Adjugates and Inverse Matrix
   - Cramer's rule
2. Diagonalization
   - Eigenvalues and Eigenvectors
   - Diagonalization
3. An Application to Linear Recurrences
4. Application in Linear Dynamical System

Linear Dynamical System

If A is an $n \times n$ matrix, a sequence $\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \ldots$ is called a **linear dynamical system** if $\mathbf{v}_0$ is specified and $\mathbf{v}_1, \mathbf{v}_2, \ldots$ are given by the matrix recurrence

$$\mathbf{v}_{k+1} = A\mathbf{v}_k$$

for each $k \geq 0$.
We call A the **migration** matrix of the system

Example - Evolution of the population

The bird population was described by computing the female population profile $\mathbf{v}_k = \begin{bmatrix} a_k \\ j_k \end{bmatrix}$ of the species where a_k and j_k represents the number of adults and juvenile females present k years after the initial values $a_0 = 100$ and $j_0 = 40$ were observed. The model assumes that these numbers are related by the following equation

$$a_{k+1} = 0.5a_k + 0.25j_k$$
$$j_{k+1} = 2a_k$$

We can write

$$\mathbf{v}_{k+1} = \mathbf{A}v_k \qquad \text{where} \qquad A = \begin{bmatrix} 0.5 & 0.25 \\ 2 & 0 \end{bmatrix}$$

We can now use our diagonalization techniques to determine the population profile $\mathbf{v}_k$ for all values of k in terms of the initial values.

Diagonalize A

The chacteristic polynomial of A is

$$c_A(x) = x^2 - 0.5x - 0.5 = (x-1)(x-0.5)$$

so the eigenvalues are $\lambda_1 = 1$, $\lambda_2 = 0.5$.

Gaussian elimination gives corresponding basic eigenvectors $\begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -0.25 \\ 1 \end{bmatrix}$. For convenience, we use multiplies $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$ respectively. Hence a diagonalizing matrix

$$P = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$$

and we obtain

$$P^{-1}AP = D = \begin{bmatrix} 1 & 0 \\ 0 & -0.5 \end{bmatrix}$$

Compute $\mathbf{v}_k$

Compute explicitly A^k for $k \geq 0$

$$A^k = PD^kP^{-1} = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & (-0.5)^k \end{bmatrix} \frac{1}{6} \begin{bmatrix} 4 & 1 \\ -2 & 4 \end{bmatrix}$$
$$= \frac{1}{6} \begin{bmatrix} 4 + 2(-0.5)^k & 1 - (-0.5)^k \\ 8 - 8(-0.5)^k & 2 + 4(-0.5)^k \end{bmatrix}$$

Hence we obtain

$$\begin{bmatrix} a_k \\ j_k \end{bmatrix} = \mathbf{v}_k = A^k \mathbf{v}_0 = \frac{1}{6} \begin{bmatrix} 4 + 2(-0.5)^k & 1 - (-0.5)^k \\ 8 - 8(-0.5)^k & 2 + 4(-0.5)^k \end{bmatrix} \begin{bmatrix} 100 \\ 40 \end{bmatrix}$$
$$= \frac{1}{6} \begin{bmatrix} 440 + 160(-0.5)^k \\ 880 - 640(-0.5)^k \end{bmatrix}$$

Thus we obtain

$$a_k = \frac{1}{6}\left(440 + 160(-0.5)^k\right), \qquad j_k = \frac{1}{6}\left(880 - 640(-0.5)^k\right)$$

Observe that

$$\lim_{k\to\infty} a_k = \lim_{k\to\infty} 440 + 160(-0.5)^k = 440$$

while

$$\lim_{k\to\infty} j_k = \lim_{k\to\infty} 880 - 640(-0.5)^k = 880$$

So in long term $a_k \approx 440$ while $j_k \approx 880$, that is the female population

stabilizes with approximately twice as many juveniles as adults.